

# Arthur Cancellieri Pires

Lead Data Scientist / Time Series & Anomaly Detection / Python, PyTorch

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## Professional Summary

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Lead Data Scientist with 7+ years turning industrial sensor data into machine learning systems for predictive maintenance in the mining and railway industry. Expert in Python, PyTorch, and deep learning for time series and anomaly detection: cut instrumentation-defect detection from a three-month cycle to real time at 91% precision, and flagged critical track defects weeks before scheduled maintenance. Led a 12-person multidisciplinary team from stakeholder requirements through the start of production deployment. Ph.D. in Mechanical Engineering (UNICAMP).

## Technical Skills

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**Languages:** Python, SQL, MATLAB

**ML & Deep Learning:** PyTorch, PyTorch Lightning, scikit-learn, TensorFlow, Keras, XGBoost, Statsmodels, Optuna, MLflow, PyCaret

**Data Tools:** Pandas, NumPy, SciPy, PySpark

**Databases:** PostgreSQL, MySQL

**MLOps & Cloud:** FastAPI, Docker, Docker Compose, Kubernetes, GitHub Actions, Git, AWS

**Visualization:** Matplotlib, Seaborn, Plotly, Power BI

**Methods:** time-series anomaly detection, autoencoders, Transformers, unsupervised learning, signal processing, feature engineering, ETL, data quality

## Professional Experience

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- Railway Laboratory (LAFER) – Predictive maintenance for VALE** Vitória, Brazil  
• **LEAD DATA SCIENTIST** *Aug 2024 to Present*
  - Led a 12-person multidisciplinary team building end-to-end predictive maintenance ML systems for VALE's track maintenance team, covering 900 km of track monitored by 2 instrumented railway vehicles, from stakeholder requirements through the start of production deployment.
  - Reduced instrumentation-defect detection latency from three months to real time at 91% precision using deep learning models in PyTorch and PyTorch Lightning on multivariate sensor streams, so faulty sensors no longer corrupted the data feeding the track-defect detection models.
  - Developed unsupervised Transformer autoencoders that identified critical track defects weeks before scheduled maintenance, including failure modes absent from the maintenance database.
- SENIOR DATA SCIENTIST** *Jan 2022 to Aug 2024*
  - Designed an automated data-quality framework in Python and SQL that scored the reliability of each data-collection run and filtered unreliable sensor data, replacing manual trip-by-trip screening and keeping the dataset valid across repeated pipeline versions.
  - Developed an unsupervised autoencoder in PyTorch, tuned with Optuna, that detected sensor anomalies missed by rule-based checks at about a 3% false-positive rate, preserving data quality for downstream modeling.
  - Built a baseline model of normal sensor behavior that gave the downstream track-defect detection models a consistent reference for training and evaluation.
- RESEARCH DATA SCIENTIST** *Jun 2020 to Jan 2022*
  - Developed a Python and SQL ETL pipeline that processed 20,000–40,000 sensor records per asset per day, merging multiple raw data sources into a unified analytical dataset that became the foundation for every later predictive maintenance model at VALE.
  - Built deep learning models in PyTorch that predicted vertical track irregularities from onboard sensor data with an R2 of approximately 0.98, reducing reliance on costly track-geometry inspections.
- Tribology and Railway Dynamics Lab (LabTDF) – Rail engineering for VALE** Vitória, Brazil  
• **MACHINE LEARNING INTERN** *Jan 2019 to Jun 2020*
  - Estimated hard-to-measure contact forces from onboard sensors using a simulation-validated ML pipeline (scikit-learn, Universal Mechanism), reaching an R2 of about 0.97, and the results helped convince VALE to launch a dedicated R&D project on track predictive maintenance using instrumented wagons.
  - Optimized wheel and rail wear profiles using genetic algorithms, reducing wear by 20% and improving a key safety ratio by approximately 35% without sacrificing fatigue performance.
  - Delivered wear and fatigue analyses that drove adoption of a maintenance threshold change in operations, extending asset service life by 29% (up to 50% combined with the optimized wheel profile).

## Key Projects

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- **Industrial Sensor Anomaly Detection API** – [github.com/pirao/AnomalyDetection2026](https://github.com/pirao/AnomalyDetection2026)
  - Built a production-style FastAPI service for real-time machine-fault detection that cut false positives to zero across 29 benchmark scenarios (F1 0.95, recall 0.91), up from a 0.28 F1 baseline.
  - Implemented per-group residual scoring on multivariate vibration sensors with a stateful multi-window alert engine (confirmation and cooldown).
  - Served versioned models from an MLflow registry, containerized with Docker Compose and tested with GitHub Actions CI.

## Certifications

2026 **Orchestrating Workflows for GenAI Applications**, DeepLearning.AI (with Astronomer) – *in progress*2026 **AI Agents and Workflow Automation with n8n**, Hashtag Treinamentos (Brazil) – *in progress*

2024 TensorFlow 2 for Deep Learning Specialization, Imperial College London (Coursera)

2024 Probabilistic Deep Learning with TensorFlow 2, Imperial College London (Coursera)

## Education

- **State University of Campinas (UNICAMP)**  
PH.D. IN MECHANICAL ENGINEERING  
Campinas, Brazil  
*Sep 2026*
- **State University of Campinas (UNICAMP)**  
M.SC. IN MECHANICAL ENGINEERING  
Campinas, Brazil  
*Jan 2022*
- **Federal University of Espírito Santo (UFES)**  
B.SC. IN MECHANICAL ENGINEERING  
Vitória, Brazil  
*May 2020*

## Languages

Portuguese (Native), English (C1 - Fluent), Italian (A2 - Basic)

## Publications & Conferences

## SELECTED PUBLICATIONS

2026 **A practical approach for reliable track defect data from instrumented railway vehicles**, Heavy Rail (first author)

2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence (first author)

2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence (co-author)

2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing (first author)

**Conference speaker:** International Conference on Railway Technology (Budapest, 2026), International Heavy Haul Association (IHHA) Conference (Colorado Springs, 2025 and Rio de Janeiro, 2023).